Brittle Star-Inspired Damage Robustness with Multi-Agent Reinforcement Learning

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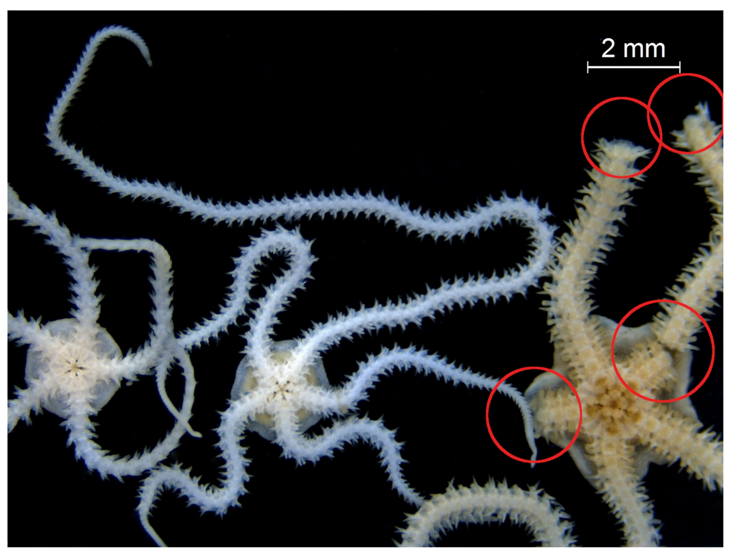
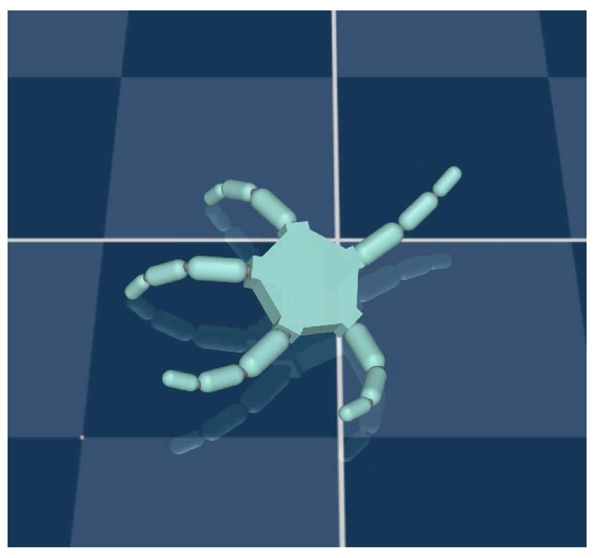
1 Damage Robustness

Real-world robots might sustain **damage** in operation. Nature already has solutions to this problem, e.g. in the form of **brittle stars**: sea creatures that are close relatives to starfish, but that can deal with the loss of limbs by **adapting their locomotion** and even regeneration.



Two intact and one damaged brittle stars.



Brittle star-inspired robot in a simulated environment.

brittle star damage robustness $\xrightarrow{?}$ damage-robust robots

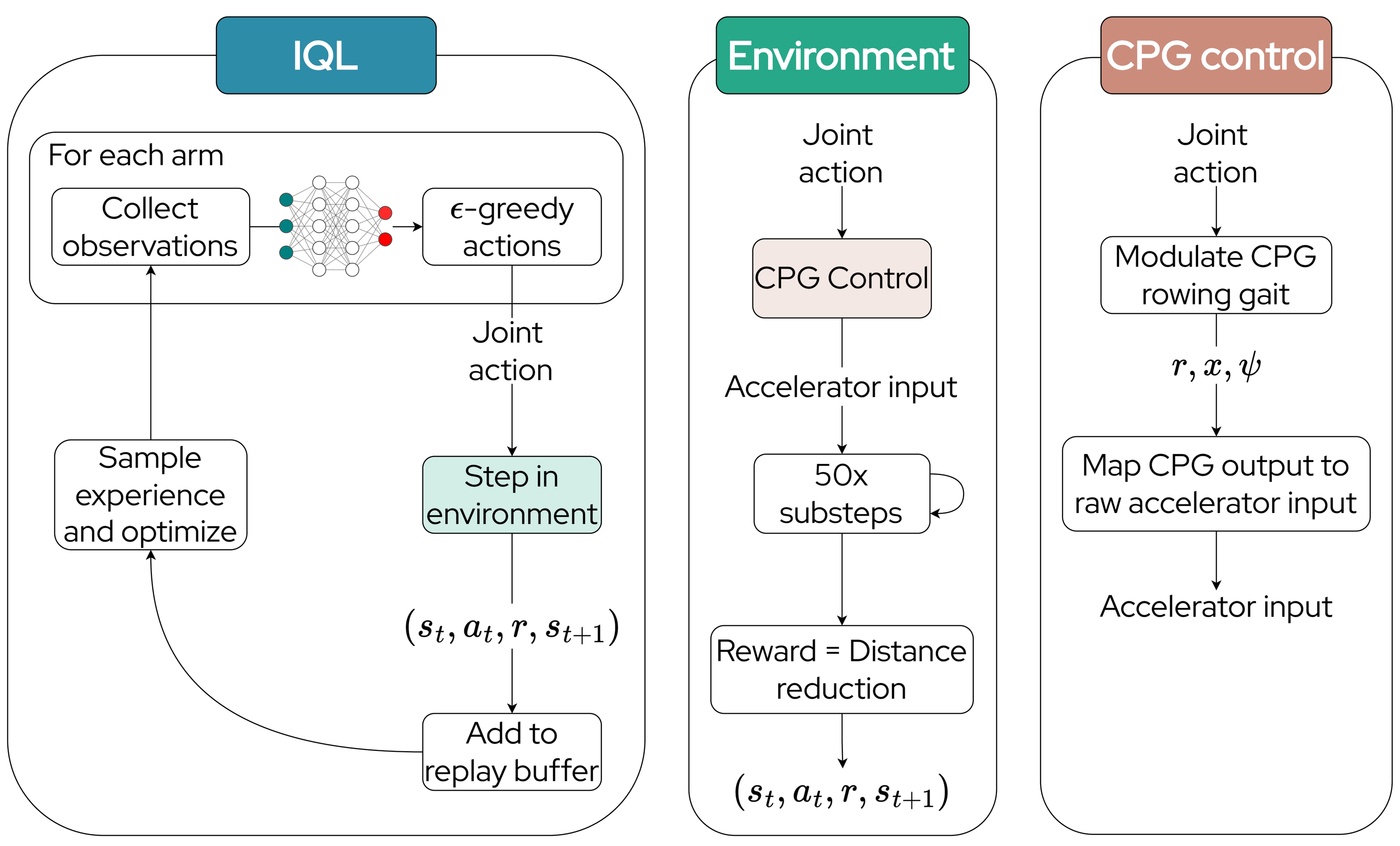
2 Research Goal

Build damage-robust robots by learning brittle star-inspired locomotion gaits with the help of central pattern generators (CPGs) and multi-agent reinforcement learning (MARL).

3 Methodology

CPGs are a **mathematical model for representing natural rythmic motions** without sensory feedback. We use them in a **MARL** pipeline where the CPG gaits are the chosen actions. The idea of MARL is to have **multiple agents interact independently** in the same environment.

We make use of **Independent Q-Learning** (IQL), which is an adaption of Q-Learning in which each agent treats the **other agents as an extra source of stochasticity**. To keep training complexity lower, agents **share their networks' parameters**.



Agents pick one of **five preconfigured CPG gaits** which allow them to switch roles from leading arm to primary or secondary rower, both left and right. Damage is introduced by **zeroing out actuator inputs for one arm** at a random point in time during an episode.

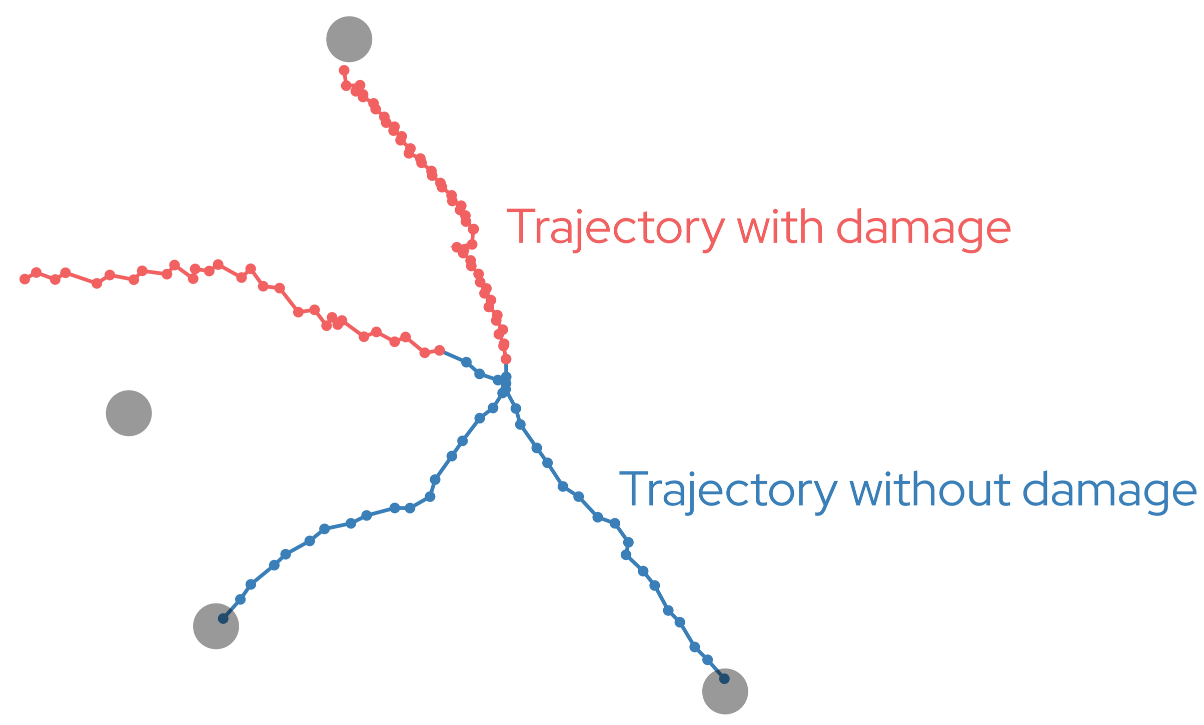
4 Results

When training with and without damage, we plotted the **number of parallel environments that reached the target in time** during each episode:

Achieving similar performance with damage robustness requires more than twice as much time

When evaluating these models, we can inspect the trajectories for two undamaged runs and two damaged runs:

Trajectories taken by the Brittle Star



5 Conclusions

1. Training to walk to a target **on a larger distance** requires more and longer training episodes.
2. Training **with damage** takes more than twice as long as training without damage to obtain similar results. Including damage during training makes the **problem more complex** and requires more training.

6 Future work

- Explore more complex locomotion gates, e.g. without CPGs
- Trying out other MARL algorithms, e.g. policy gradient

7 References

1. Albrecht, S. V., Christianos, F., & Schäfer, L. (2024). Multi-agent reinforcement learning: Foundations and modern approaches. MIT Press.
2. Simoens, P. (2026). Reinforcement Learning. Department of Information Technology, Ghent University.